

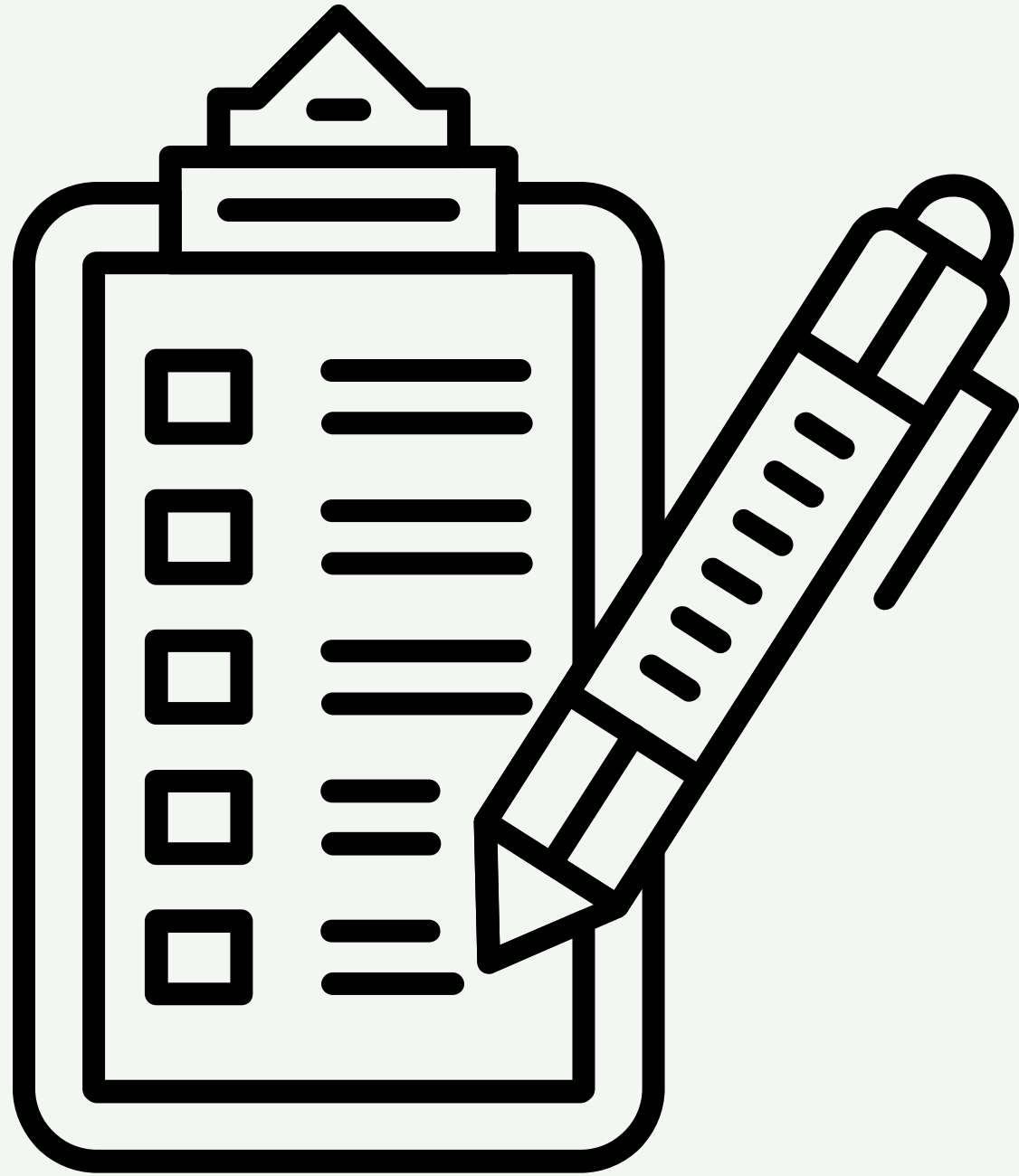


Remote Work Impact On Productivity and Well-being

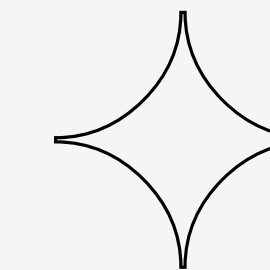
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IS 545 Spring 2026

Agenda

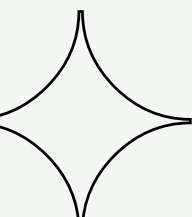


- Dataset
- Demographics
- Analysis
 - Remote work and sleep quality
 - Work environment and stress level
 - Satisfaction & Productivity
- Takeaways
- Company Recommendations



Dataset Overview

- The dataset focus on the mental health, stress level, and job satisfaction.
- The dataset is from Kaggle
- Dateset name: Remote Work & Mental Health



Data Includes:

Stress & Mental Health Factors – Simulated self-reported stress levels & conditions
Social Isolation Ratings – Employees' perception of workplace connectivity
Job Satisfaction Trends – Modeled patterns of employee satisfaction
Work Life Balance Score – Measure how well individuals manage between work and life



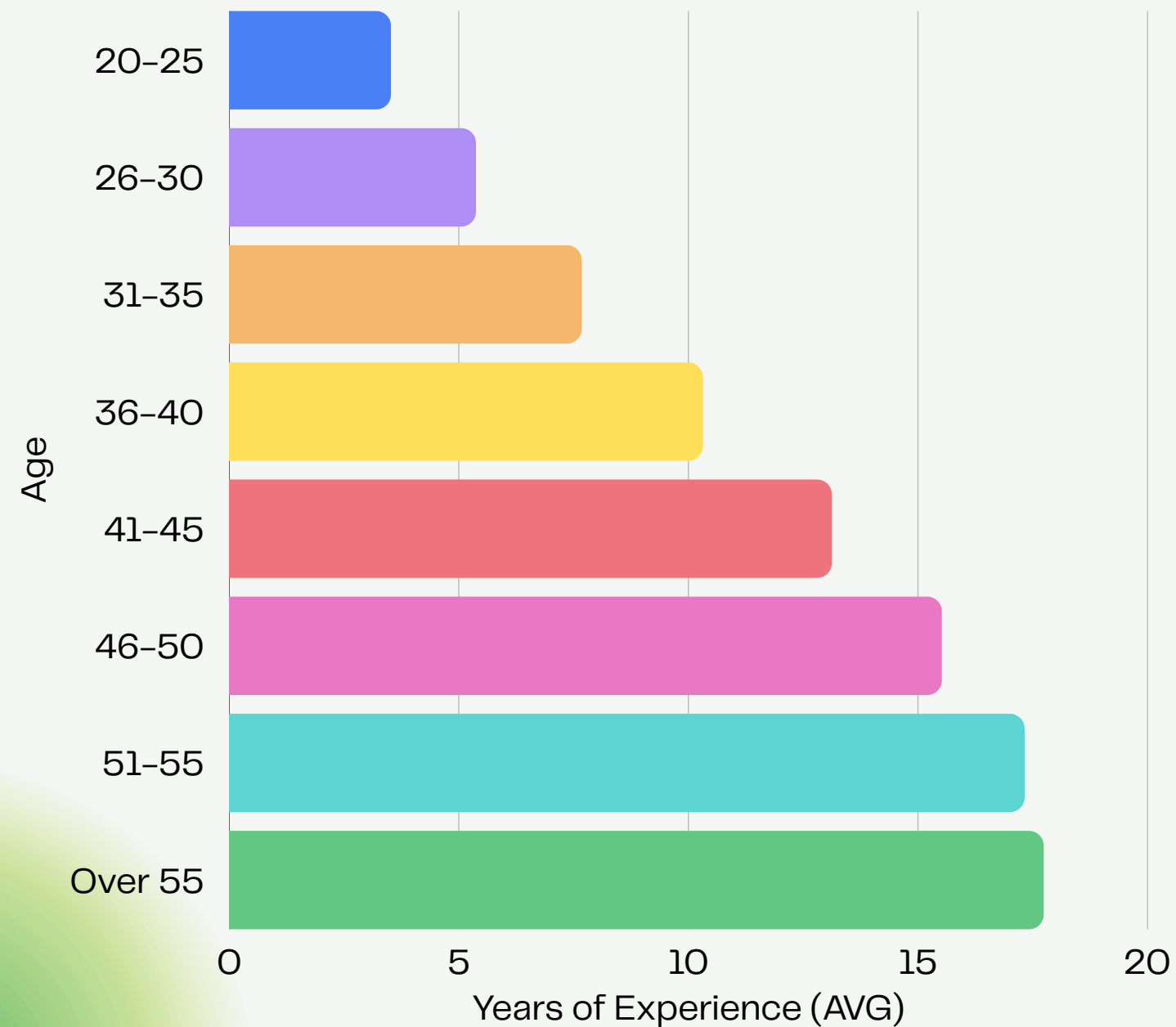
Data Cleaning & Challenges

- Excel used to clean data
- Removed any NULL responses
- Removed participants whose years of experience did not match their age



Data Validation

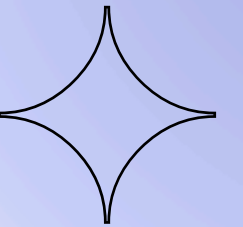
Age and Years of Experience



- Average experience rises from **early-career (20–25)** to **55+**.
- The trend is smooth and predictable, showing a **strong** age–experience **relationship**.

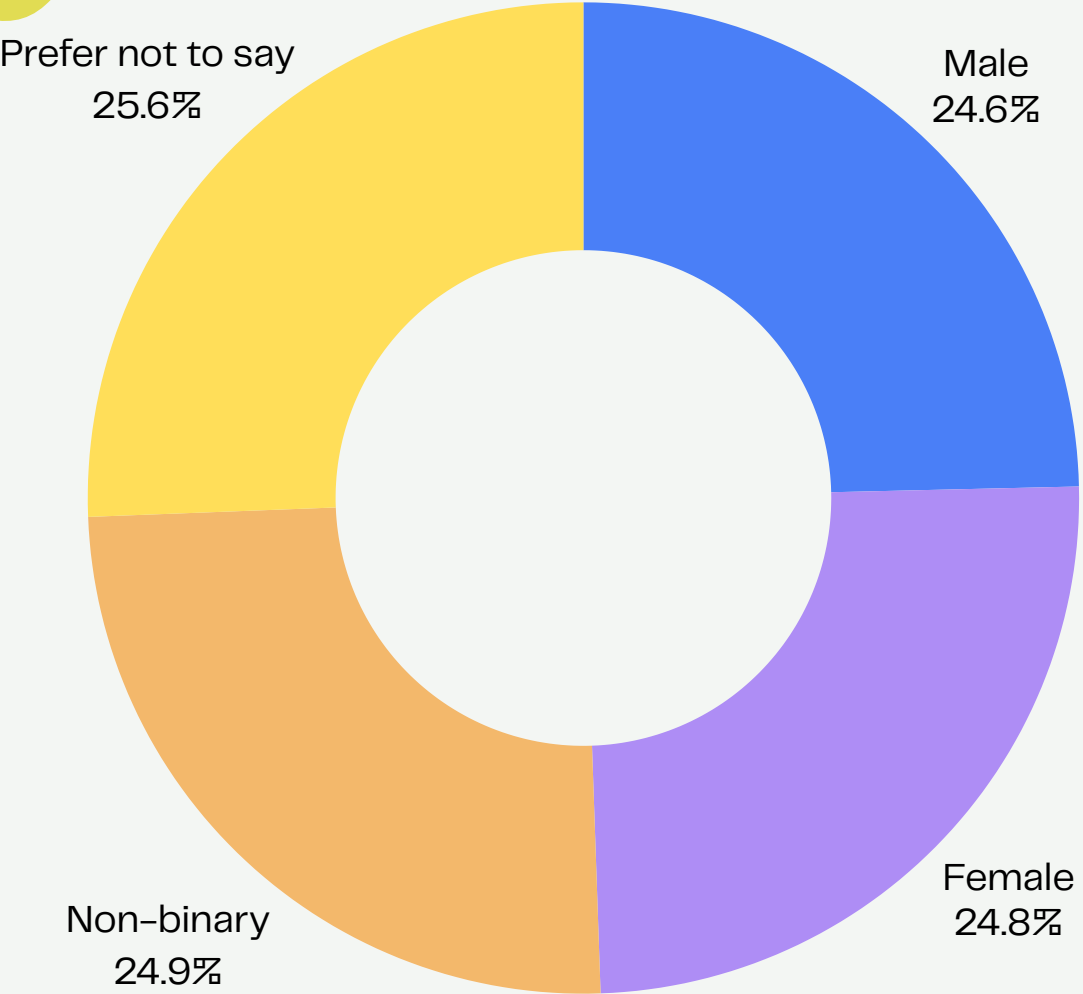
Why Calculate This?

- Helps to confirm that our data is **reliable**
- Makes sure our data is **consistent** with the **real world**
- This is an important **first step** to **certifying** this data before we continue!

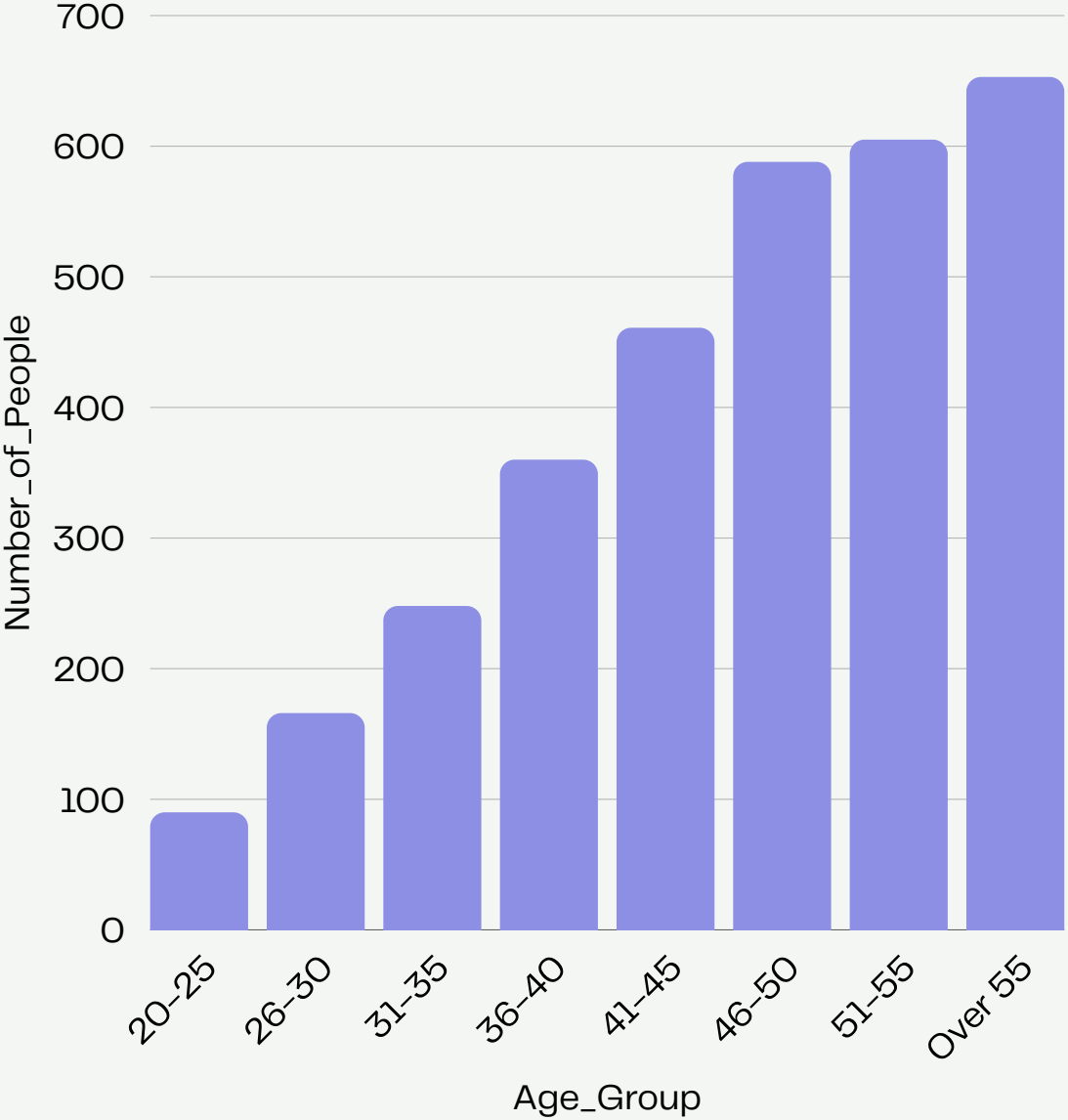


Demographics

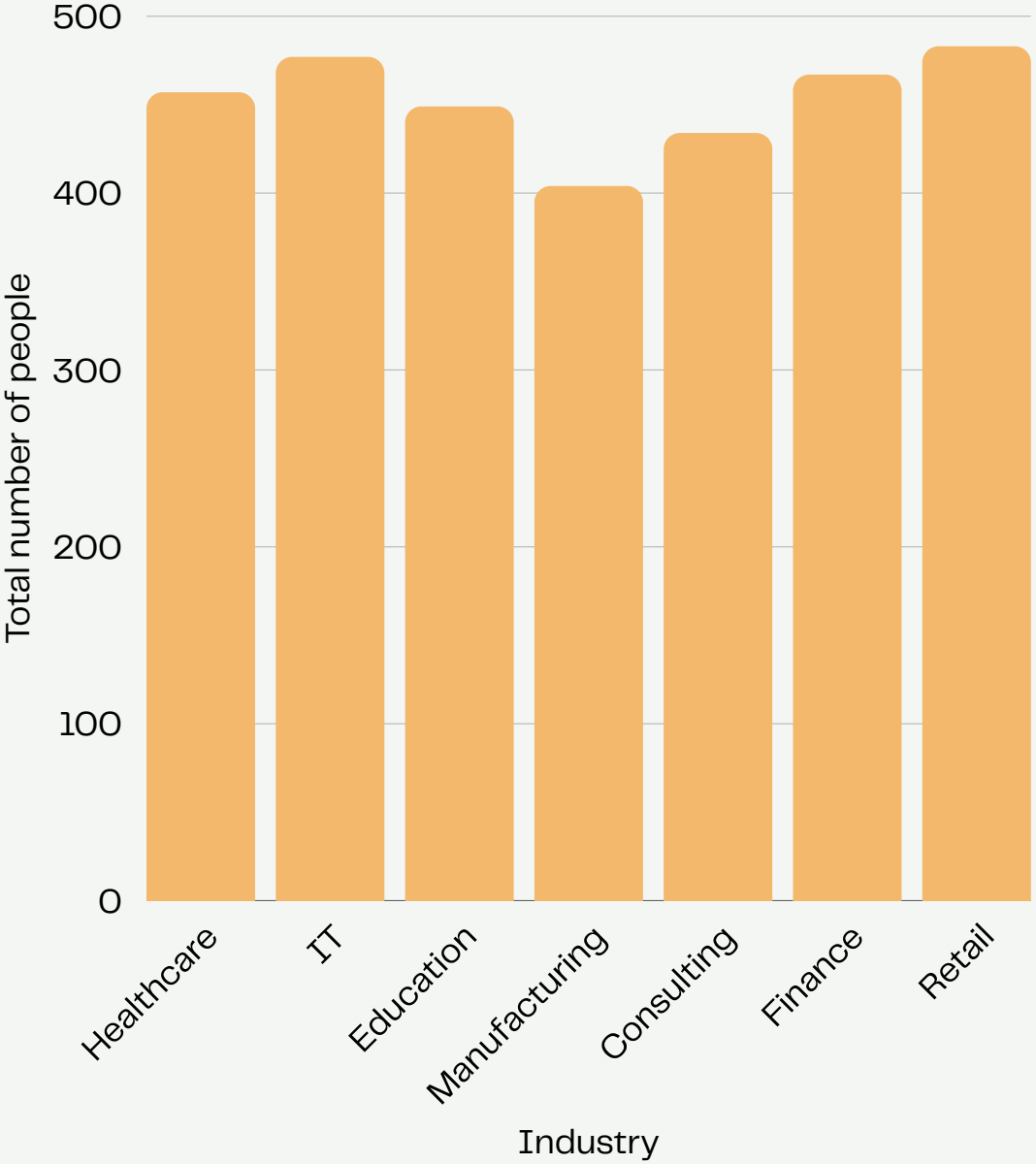
Gender



Age



Industry



Why We Will Focus on IT, Finance, and Consulting

Creating a Work-Life Balance

For the sake of cleaning up our data:

- ★ The dataset included many industries, but we narrowed our scope for **relevance**.
- ★ **IT, Finance, and Consulting** align closely with the themes of our Information Systems course.
- ★ These fields offer higher availability and likelihood of **remote** or **hybrid work**, making them ideal for analyzing work-location trends.



Experience by Gender Across Job Roles

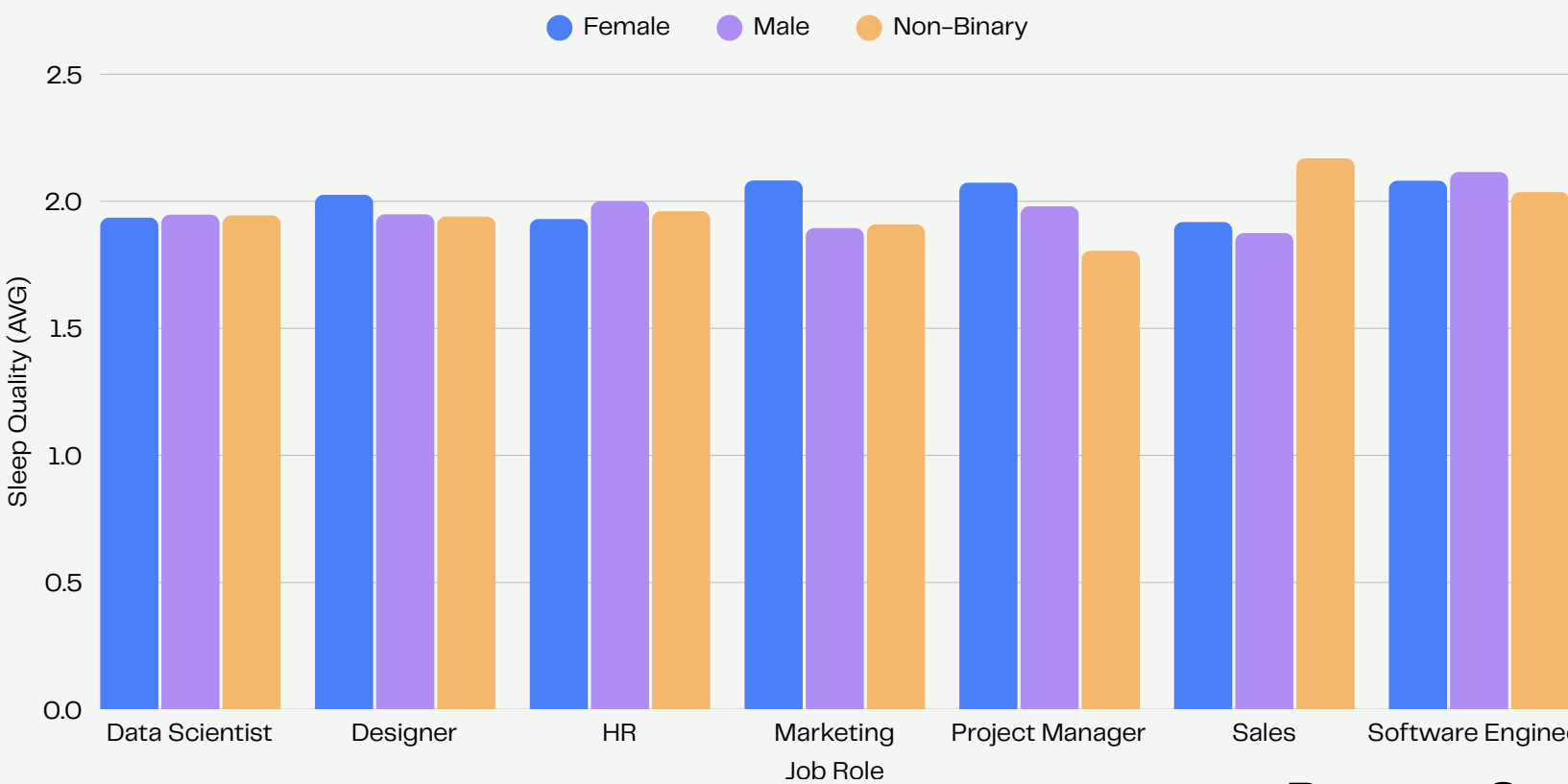
Female Male Non-Binary



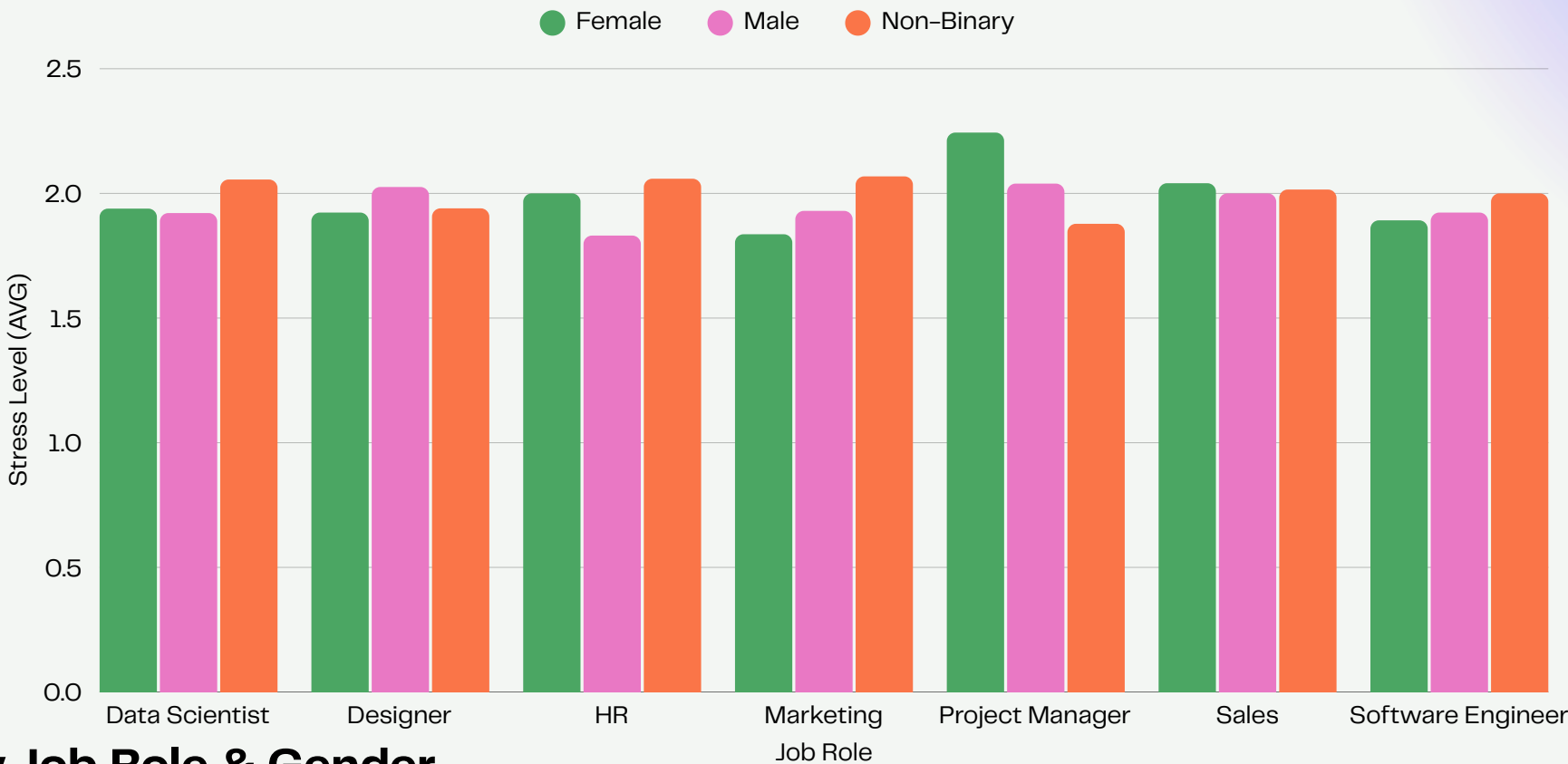
How this Ties to Sleep & Remote

Valued on a 1 – 3 scale (higher = better sleep / higher stress / higher satisfaction)

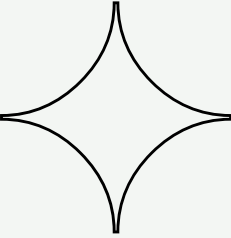
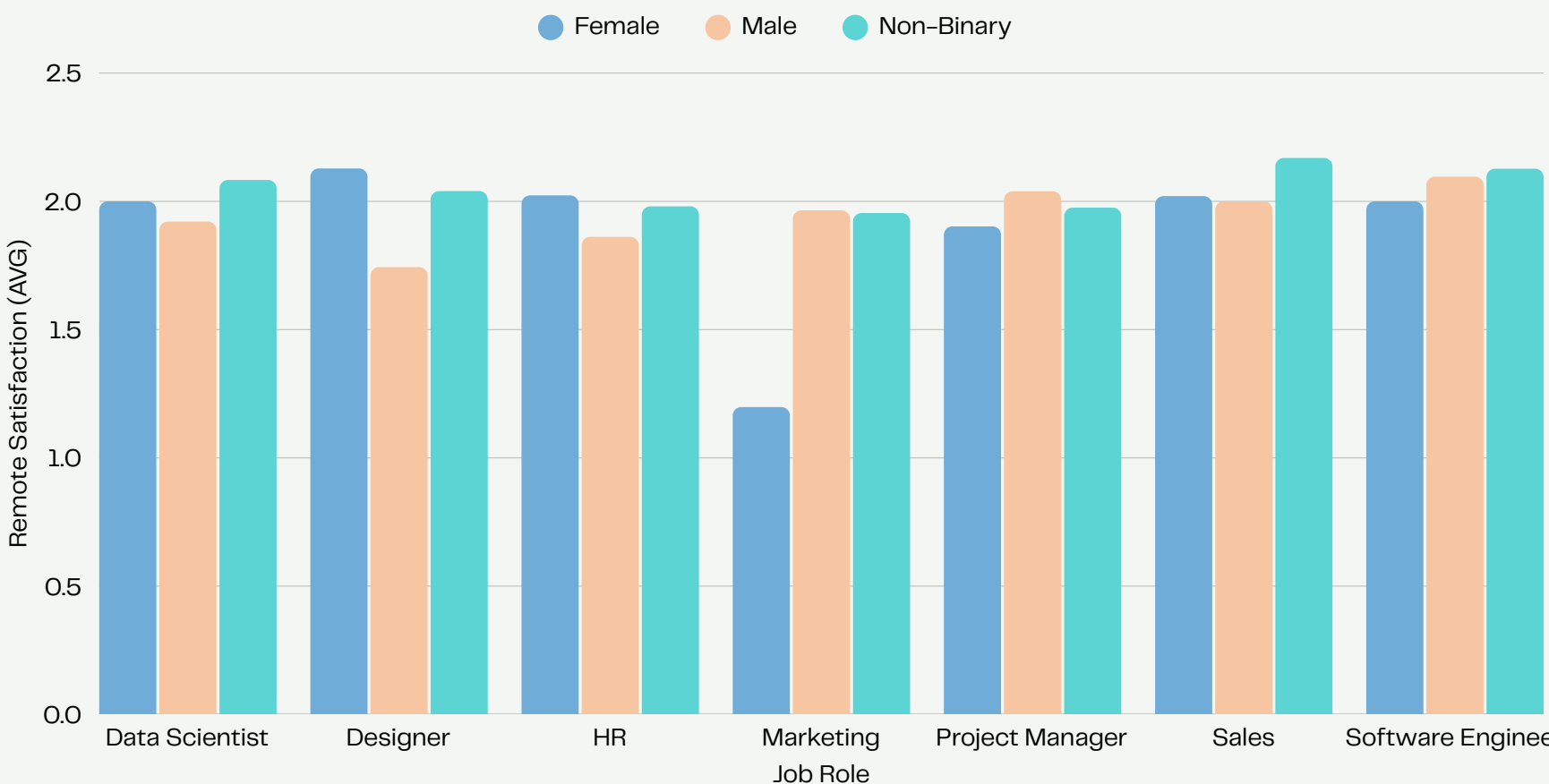
Sleep Quality by Job Role & Gender



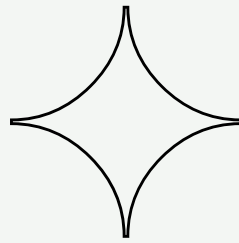
Stress Level by Job Role & Gender



Remote Satisfaction by Job Role & Gender



Summary of Insights

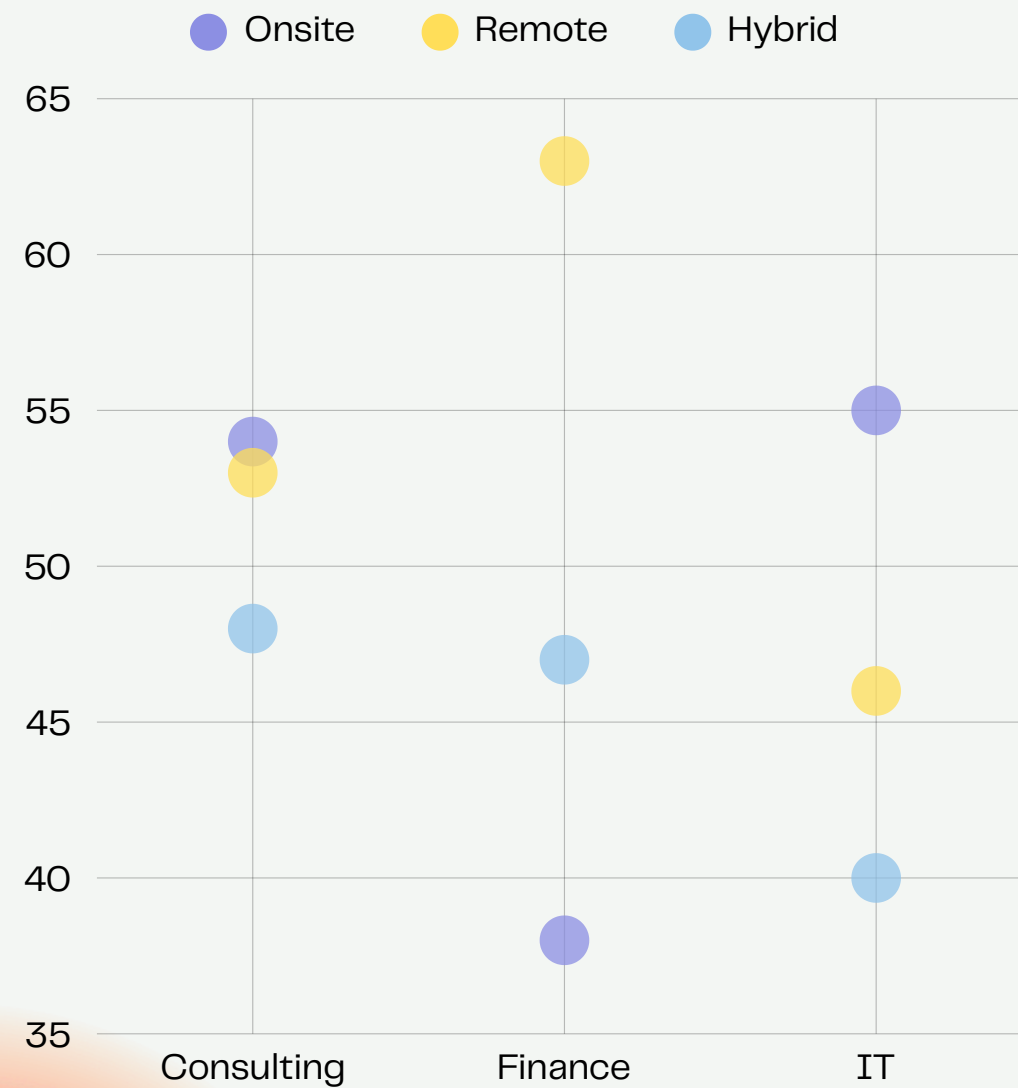


- Differences were **small** across genders.
- **Job role** had a **stronger** impact on sleep, stress, and remote satisfaction.
- **Project Managers** showed higher stress overall.
- **Software Engineers** and **Marketing** roles reported slightly **better sleep**.
- **Sales** and **Software Engineering** had higher **remote-work satisfaction**.
- Overall patterns suggest **role > gender** in shaping well-being.

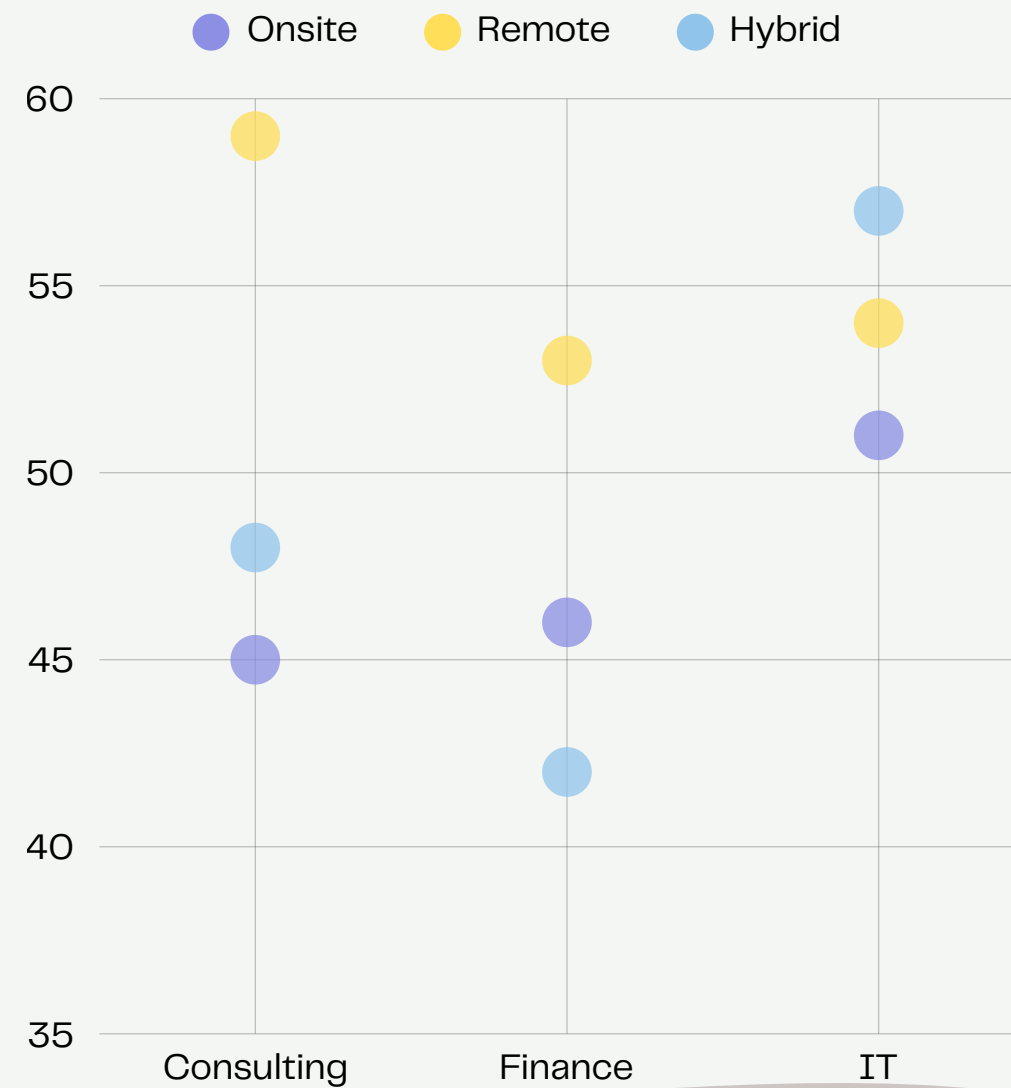
Industry Sleep Quality Analysis



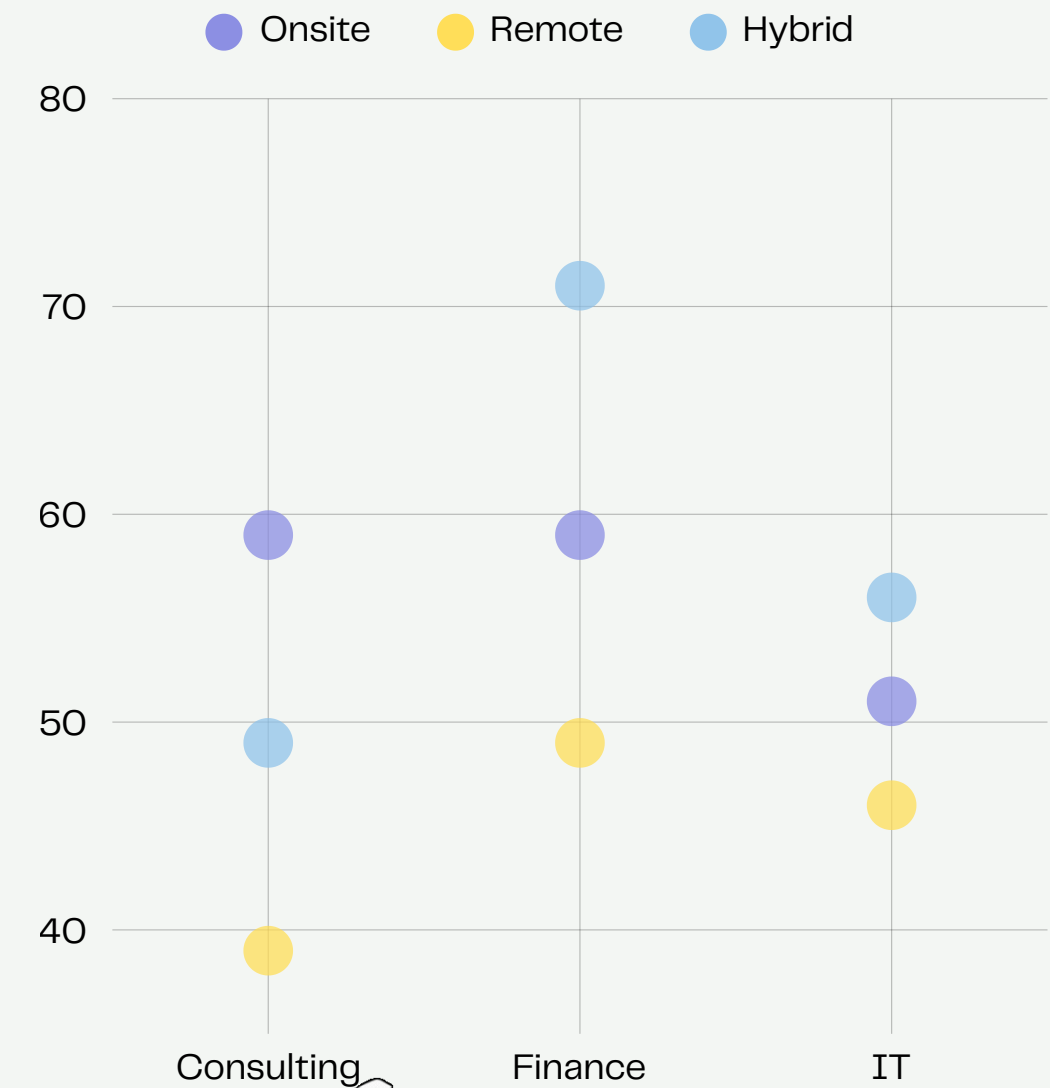
Good Sleep Quality



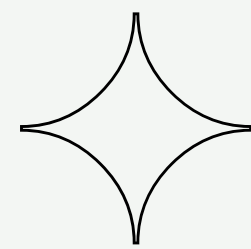
Average Sleep Quality



Poor Sleep Quality

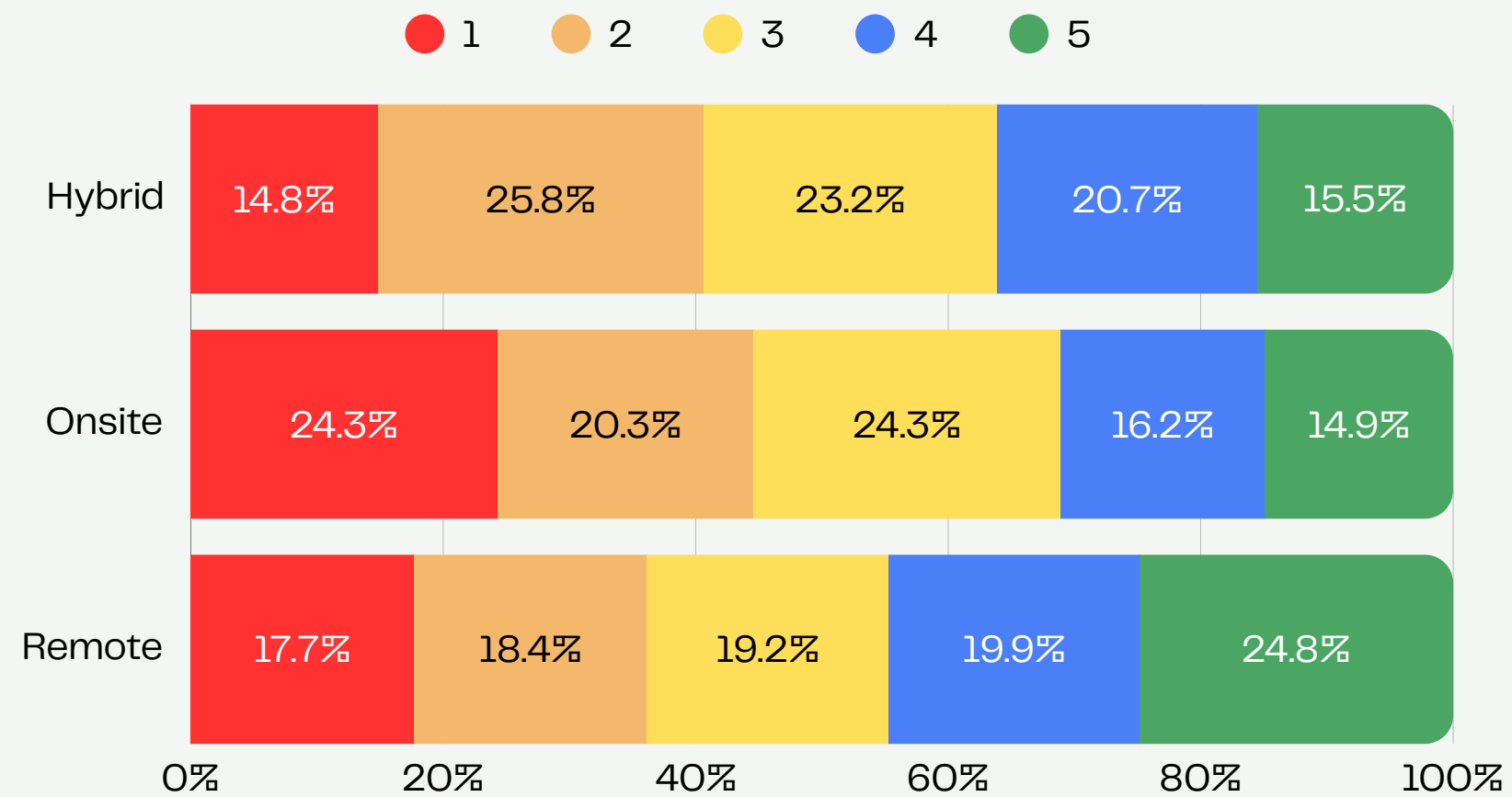


- Remote work typically results in “average” sleep quality
 - steady
 - not optimal
- Hybrid work outcomes show significant variability
 - instances of both high and low sleep quality
- Onsite work generally depends
 - often landing between the two extremes

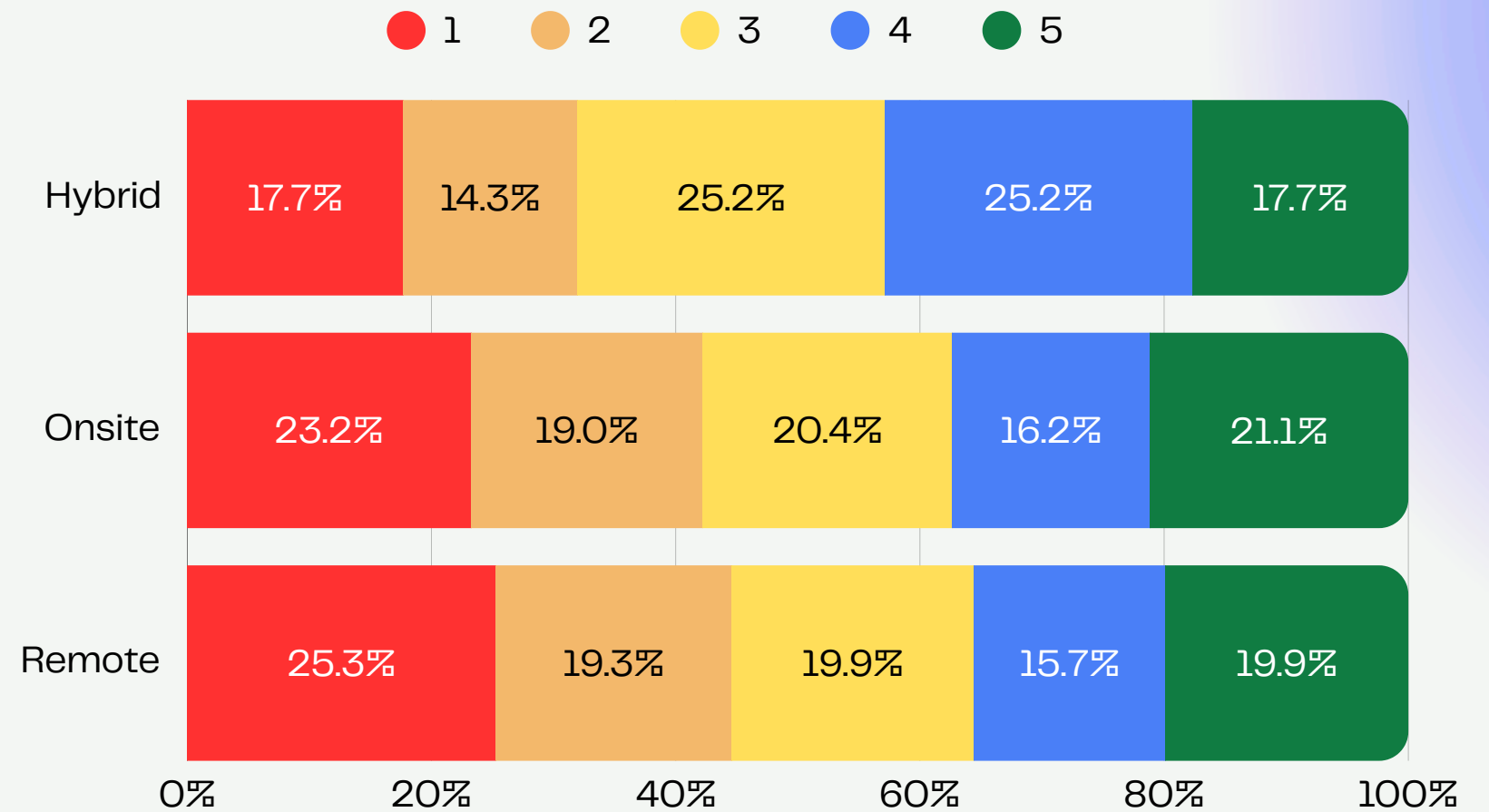


Sleep Quality and Job Satisfaction

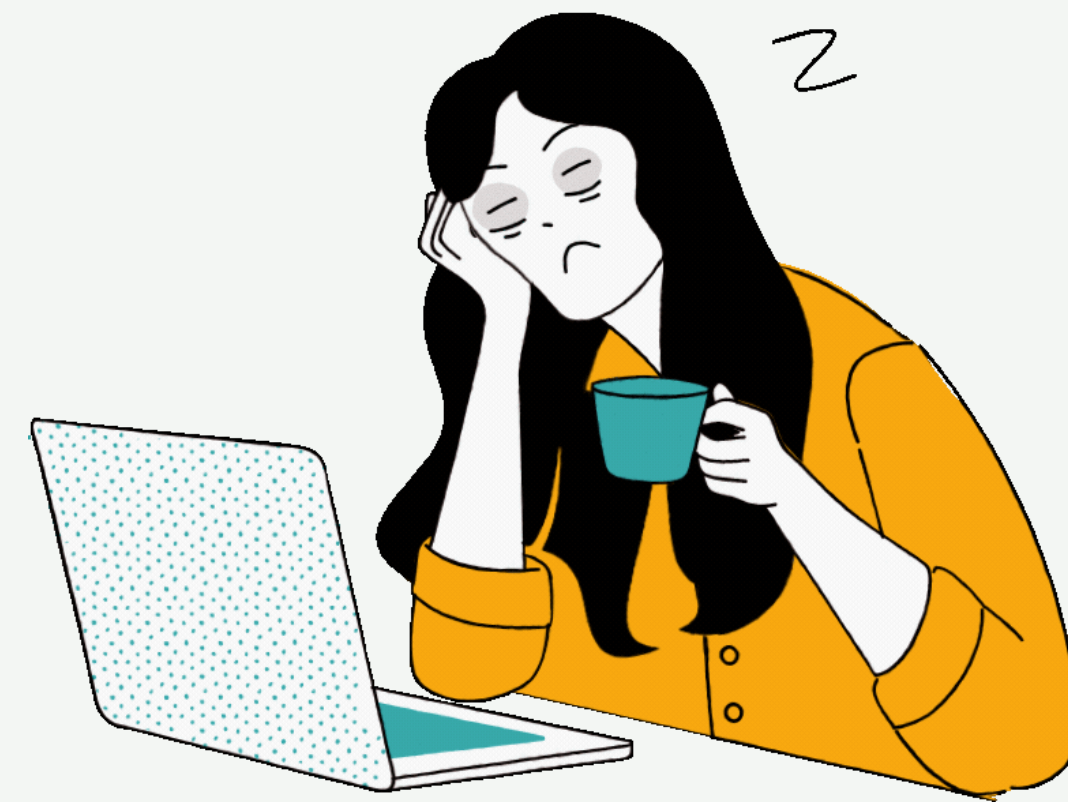
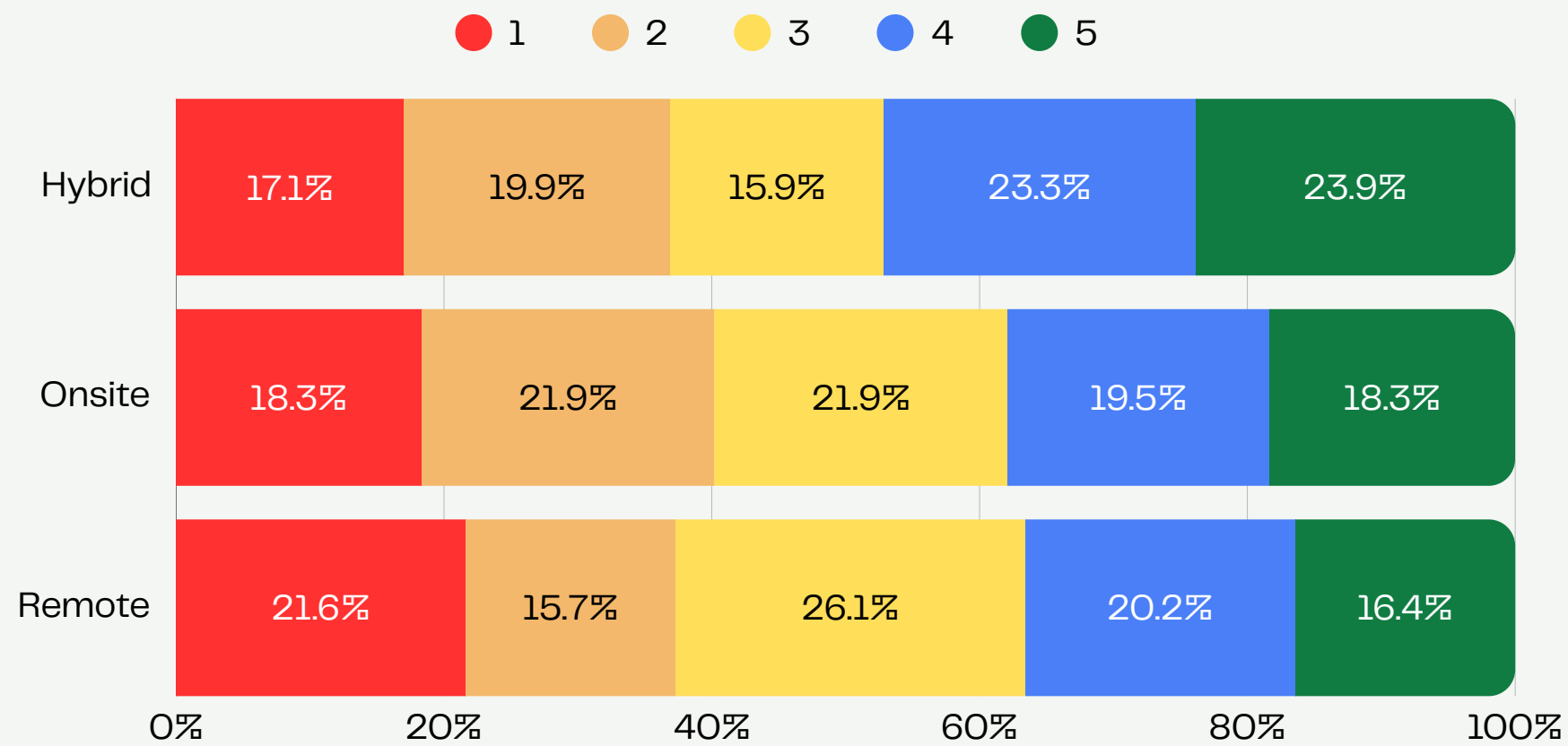
Good Sleep Quality



Average Sleep Quality



Poor Sleep Quality

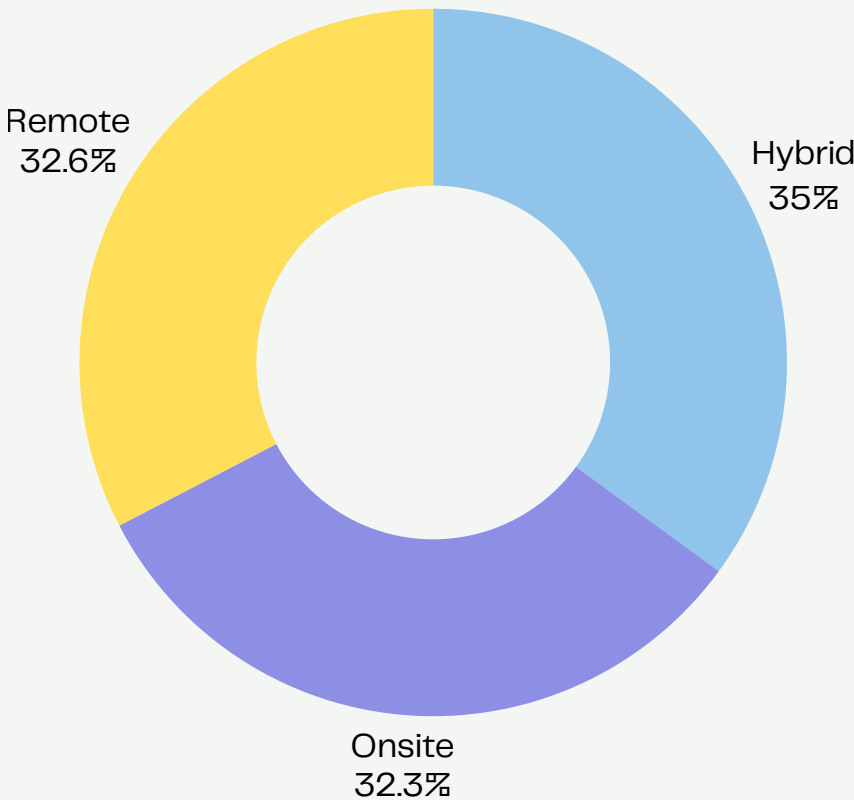


Does Remote Work Lead to Better Sleep?

WORK LOCATION
NOTICEABLE
RELATIONSHIP
WITH SLEEP
QUALITY

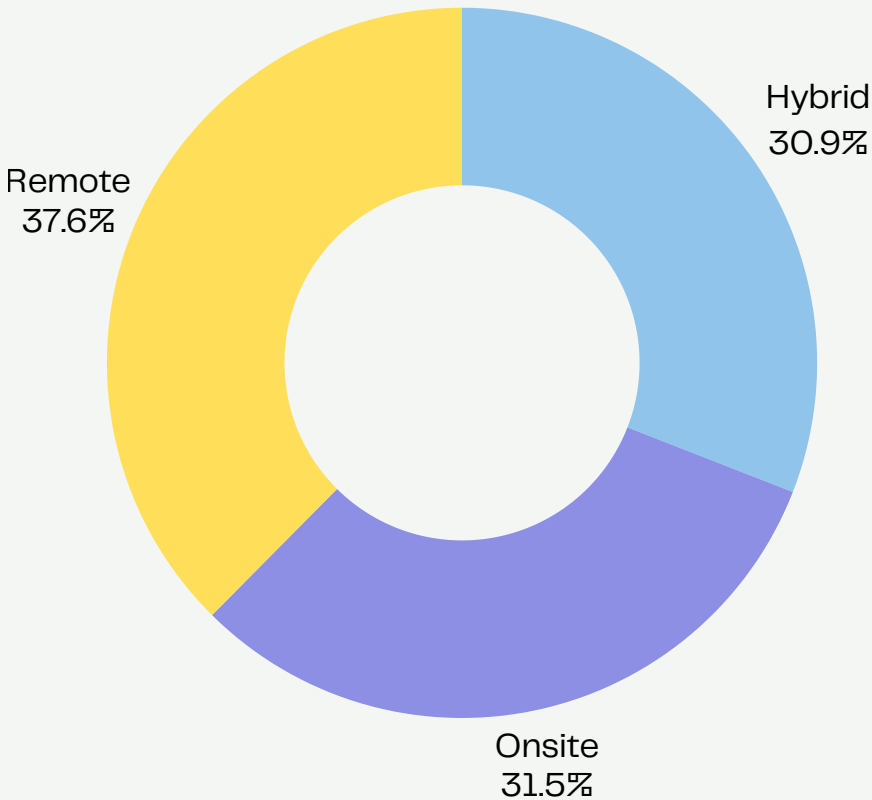
Good Sleep Quality

Hybrid Onsite Remote



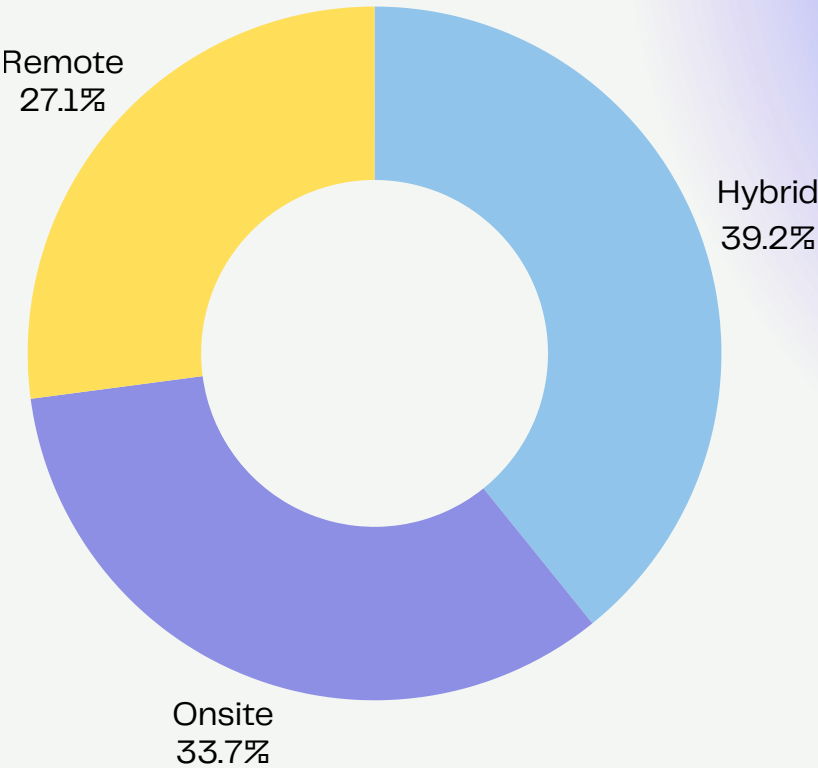
Average Sleep Quality

Hybrid Onsite Remote



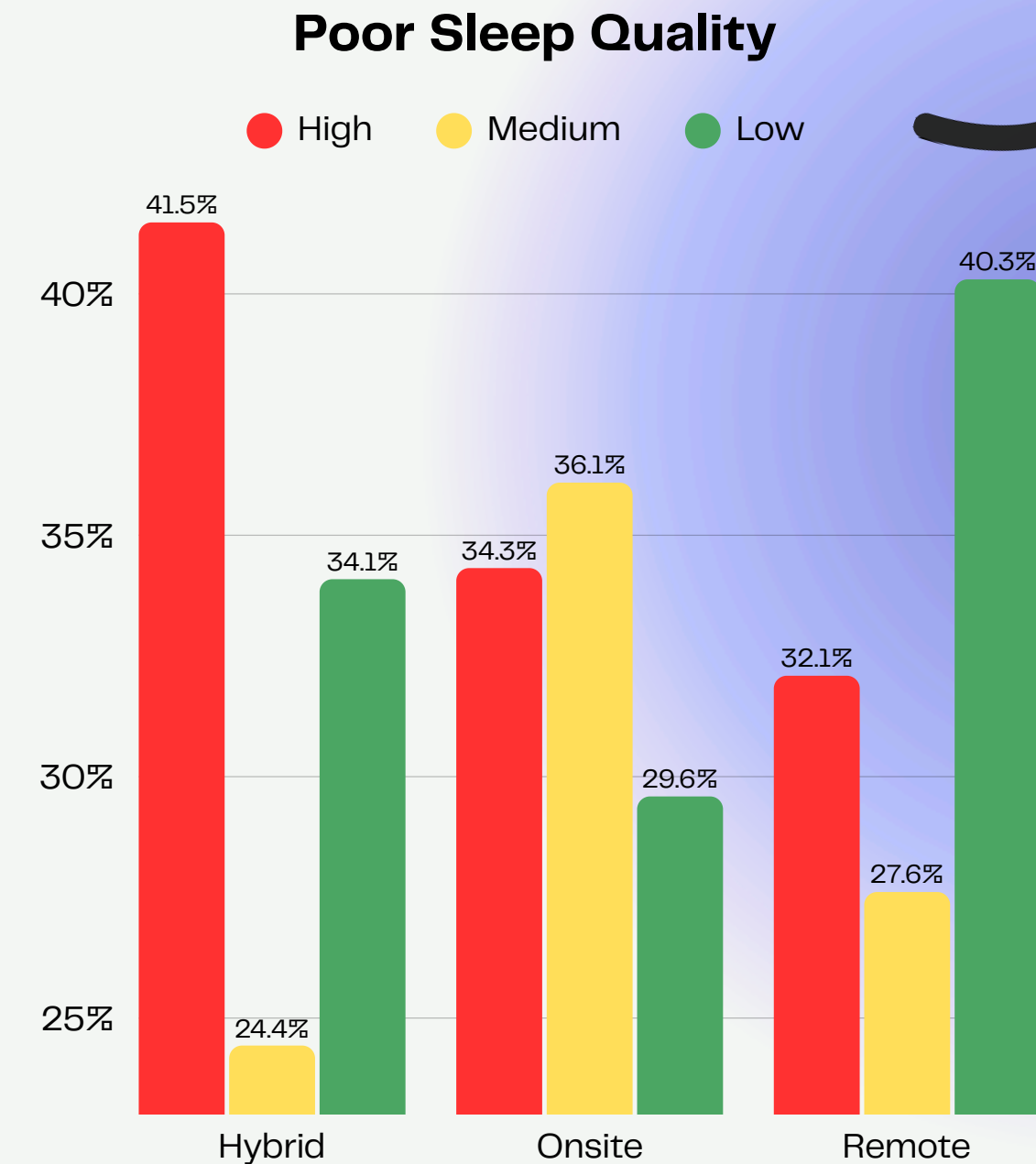
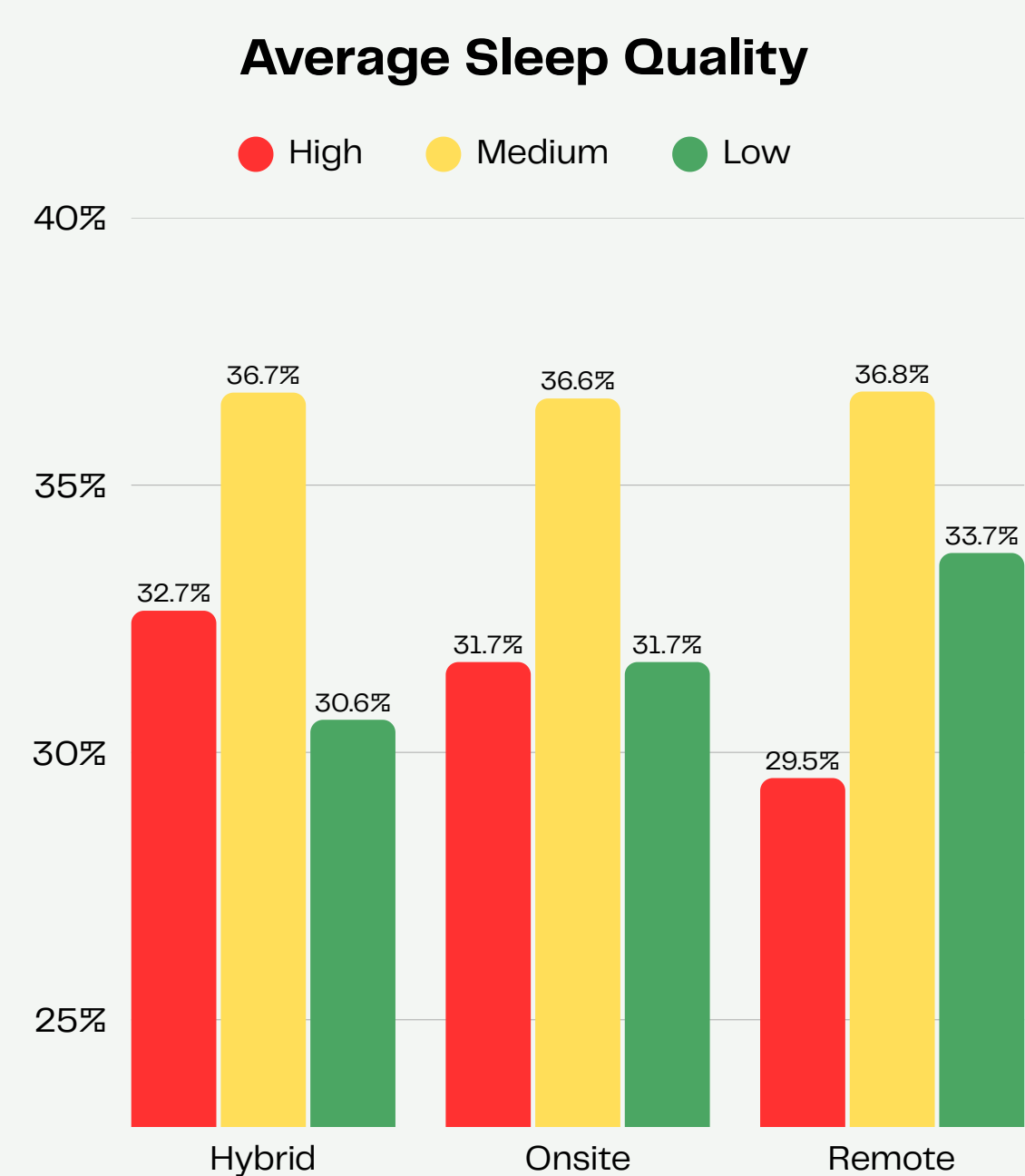
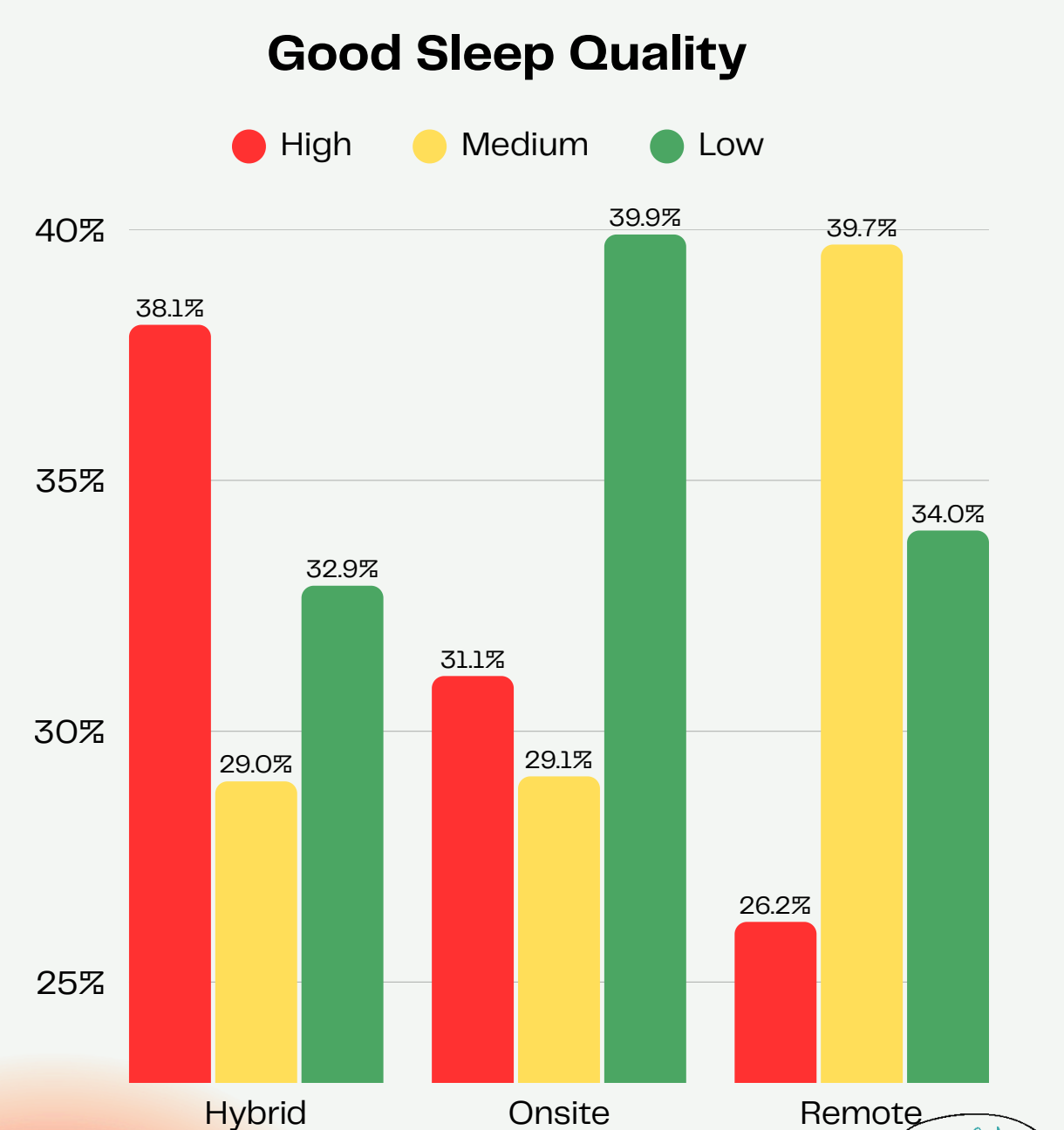
Poor Sleep Quality

Hybrid Onsite Remote



Remote work associated with **better** outcomes
hybrid with **worse**.

Finding Balance in a High-Stress Workforce

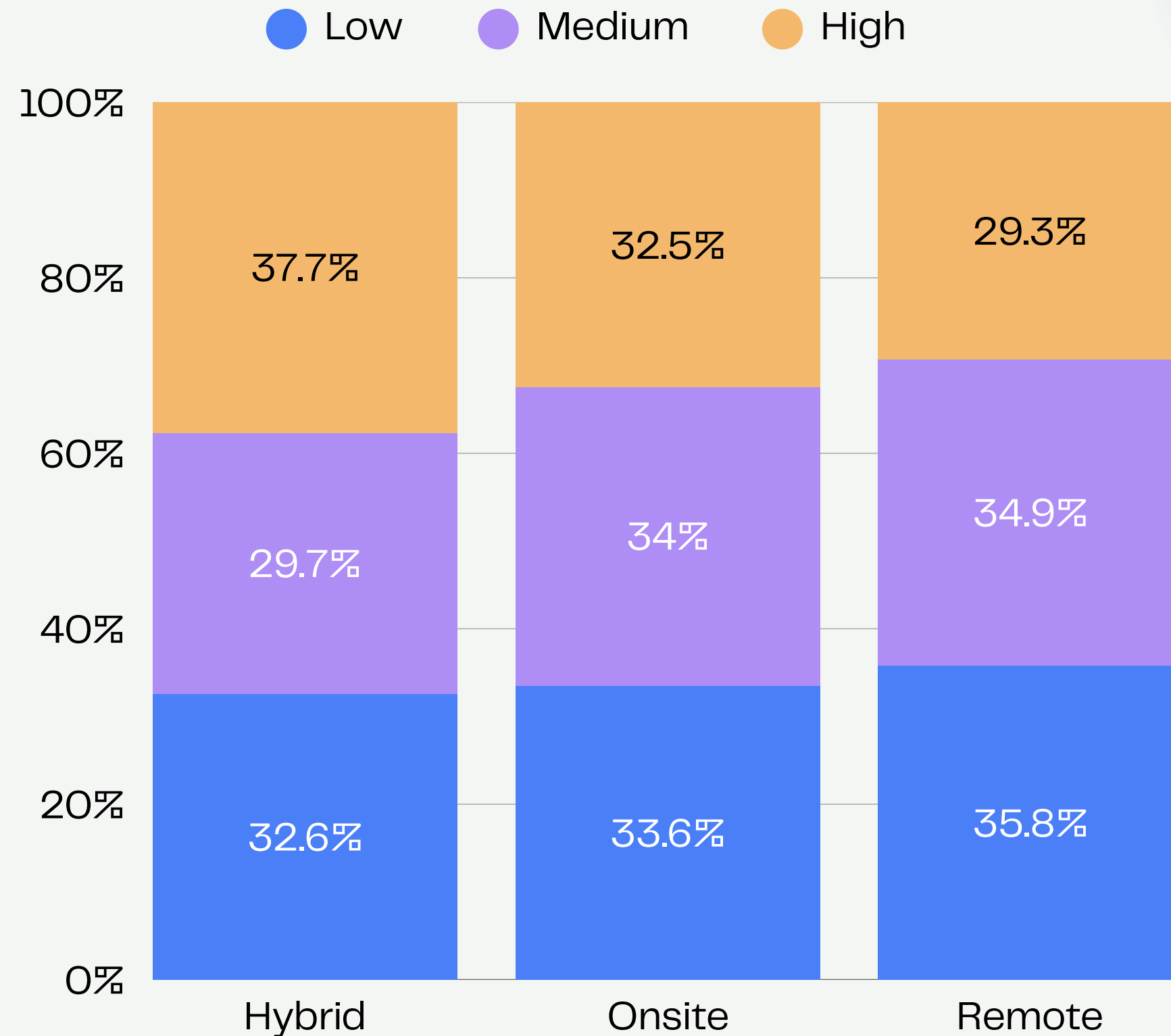


- High stress = poor sleep
- Low stress = better sleep
- Medium stress = average outcomes

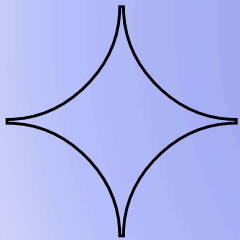
Sleep reflects the problem = stress is the cause.
So what's driving the stress?

Does Stress Related to Work Location

Stress and Work Location

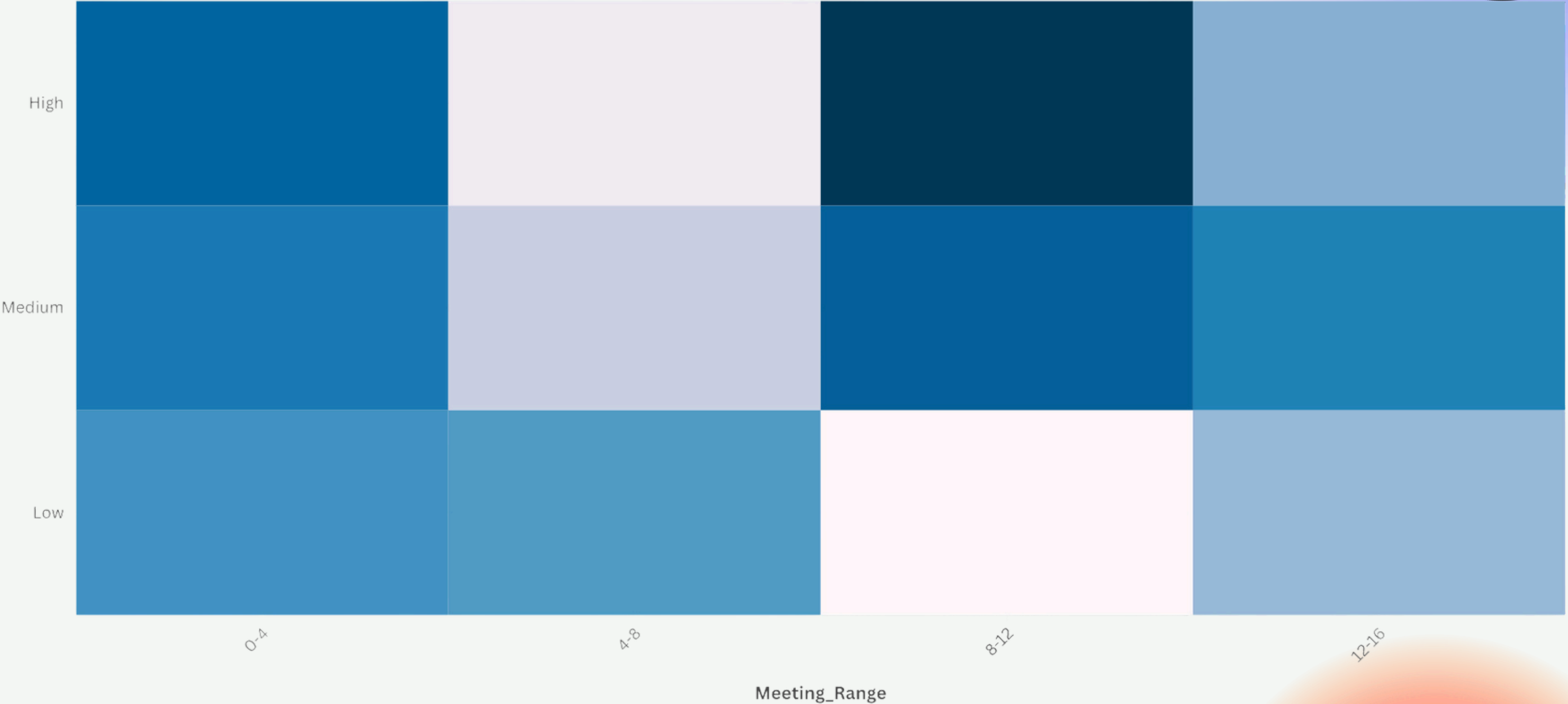


Numbers of Meeting vs Stress



Employee_Count 240 285

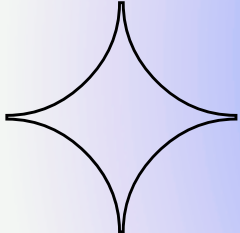
Stress_Level



- Tipping point at 8–12 meetings per week
- Less people shown in stress after 12 meetings per week

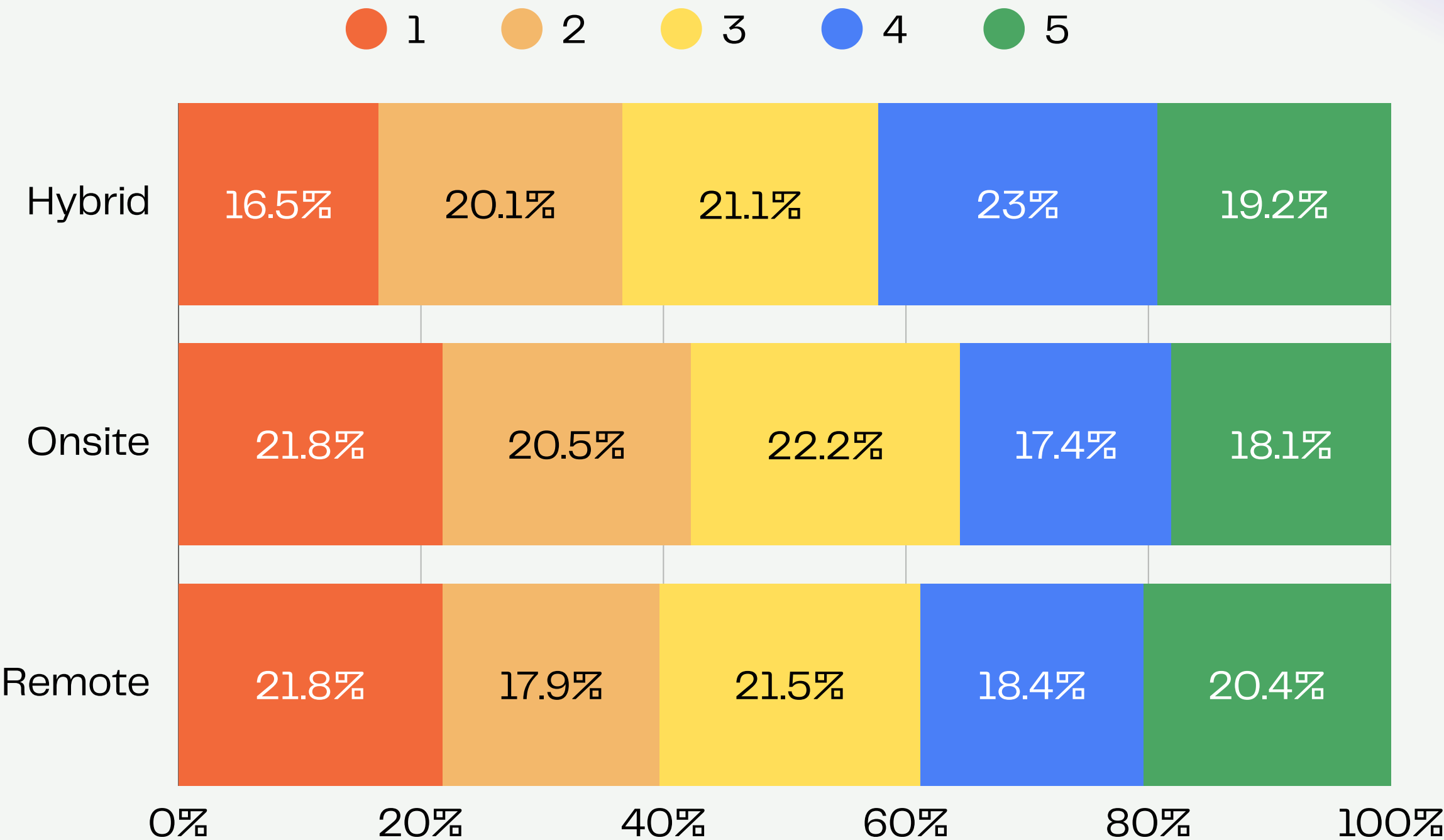


Work-Life –Balance With Remote Work



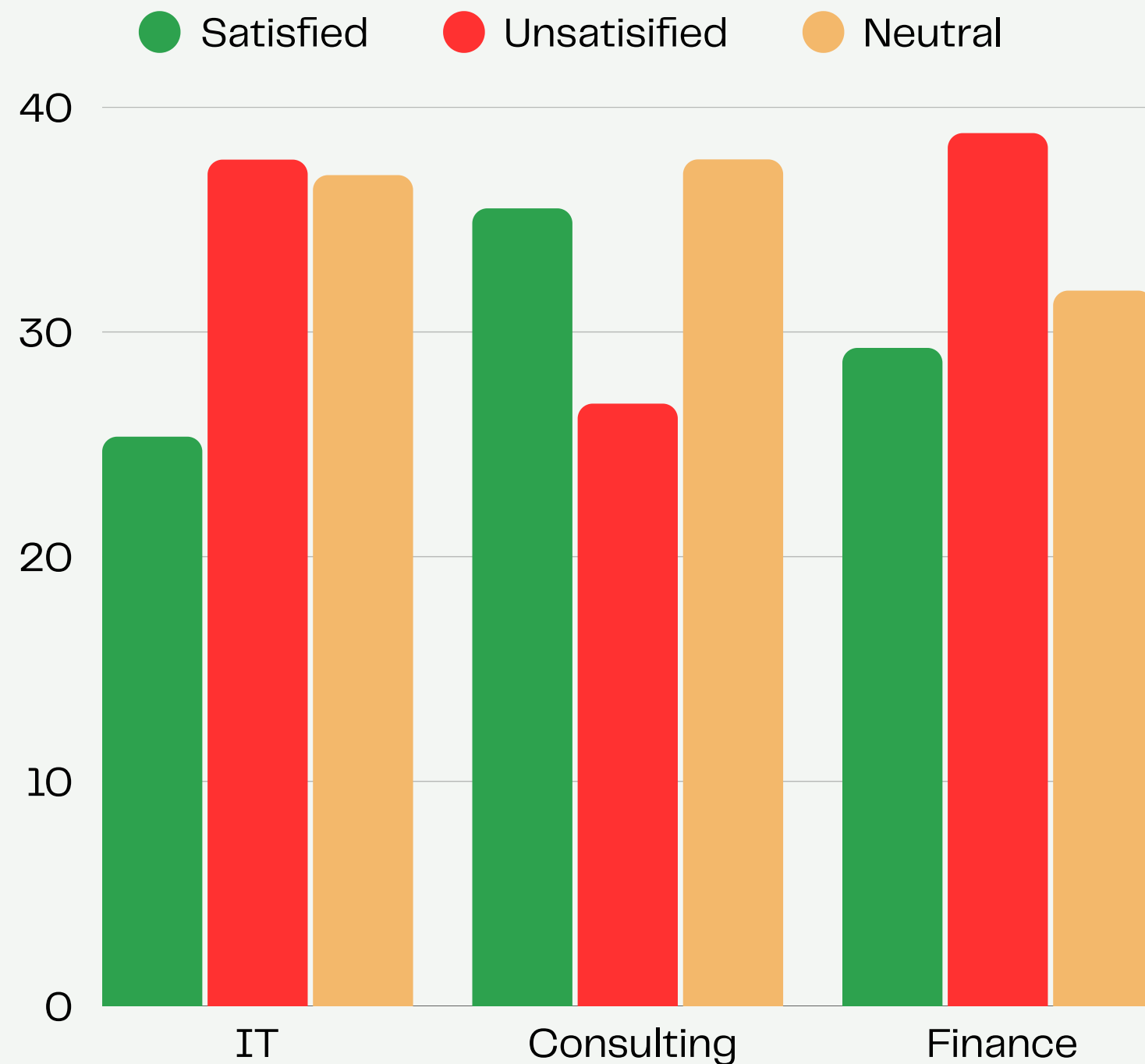
Work – Life Balance Distribution Across Location

- Hybrid worker seems to be most “stable”
- Both onsite and remote are showing higher percentage in poor work-life balance rating



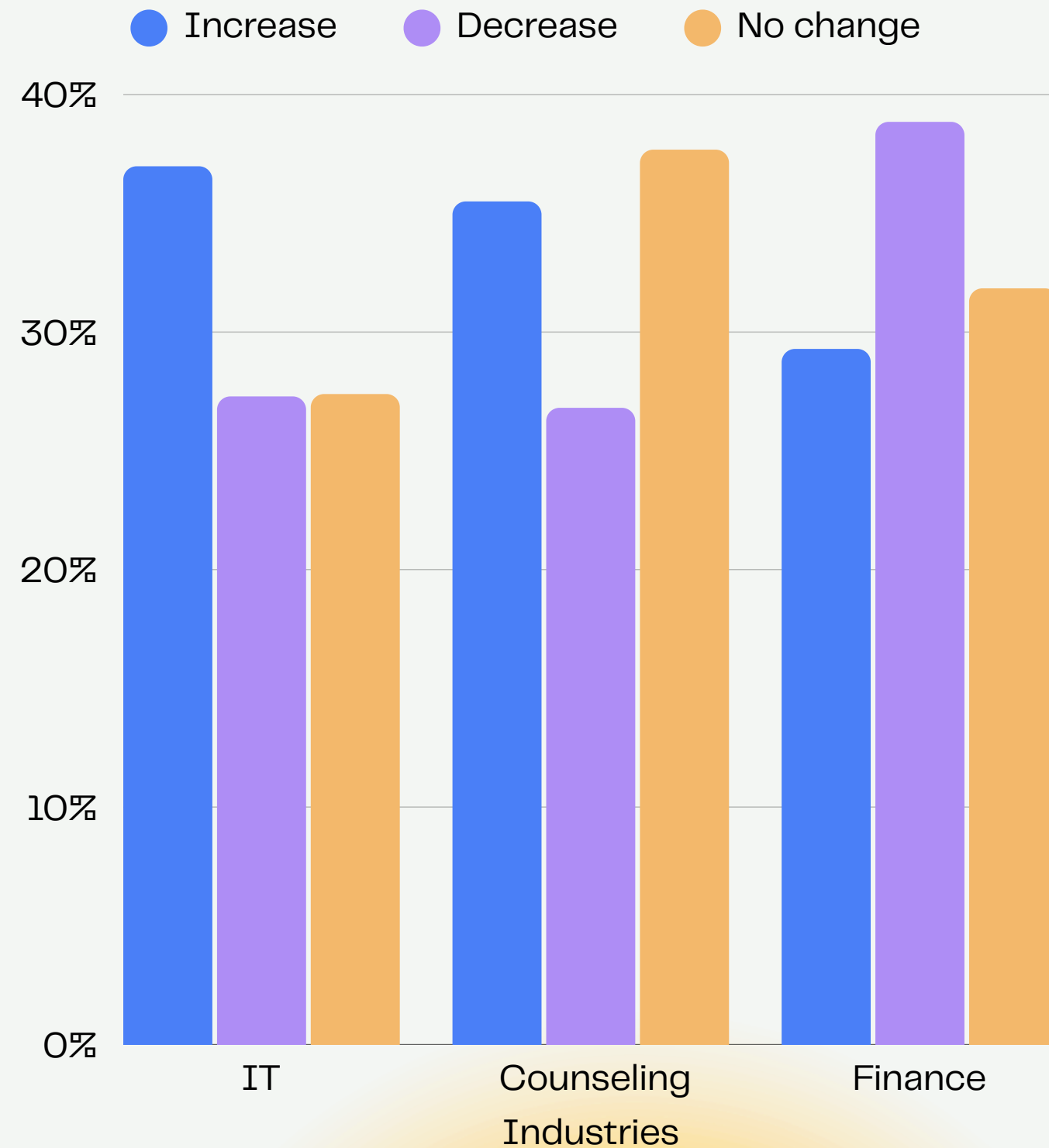
Satisfaction rates by industry

Remote workers satisfaction rates



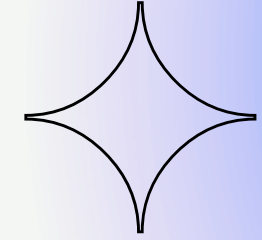
- IT shows the **lowest** satisfaction rate (25%) and **high** dissatisfaction (37.5%)
- Consulting has the **highest** satisfaction rates (35.5%) among the 3 industries
- Finance workers are the **most dissatisfied** (38.8%)
- Neutral responses are similarly high across the three industries

Productivity rates by industry



- IT workers report the **highest productivity increase** (36.9%)
- Consulting shows the **highest drop** in productivity (45.6%)
- Finance has the **most balanced distribution**

Data Usage and Limitation



Data Usage:

- HR
- Managers or Leadership
- Companies (especially IT, Finance, and Consulting)



Limitation:

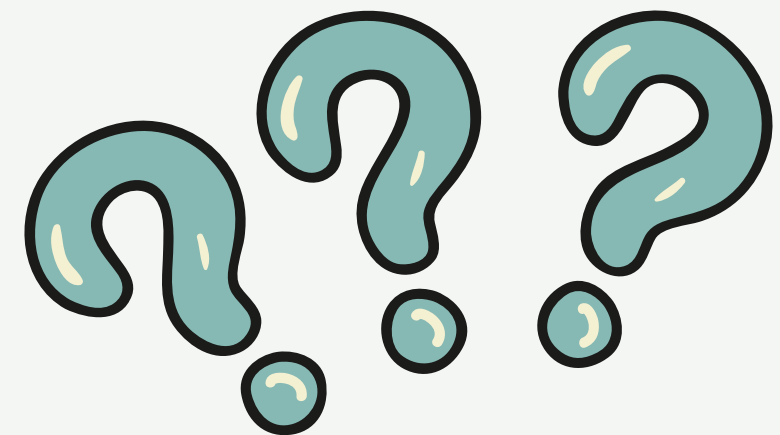
- Synthetic data
- Data could include more metric such as education, salary, workplace condition or commute time to make more informed conclusions
- Some metrics are quite vague (e.g. Work-level balance score), hard to interpret

Takeways

- Role matters more than gender
- Stress consistently drives negative outcomes
- Meetings have a tipping point where they hurt performance
- Productivity and satisfaction vary by industry

Each work model shows a different pattern, shaped by stress, workload, and individual work styles.





Q&A

